RFC-019: Migrate Project Atlas to Vector Database

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Summary

Project Atlas, our internal recommendation engine, currently runs on a single Postgres instance handling both relational data and similarity search via a hand-rolled cosine-distance query. As our catalog has grown past 2 million items, p95 query latency has climbed from 80ms to over 900ms. This RFC proposes migrating the similarity-search component of Project Atlas to a dedicated vector database.

Problem

The current implementation stores item embeddings as a Postgres array column and computes similarity with a custom SQL function. This does not scale: query latency grows linearly with catalog size, and the lack of an approximate-nearest-neighbor index means every query is a full table scan.

Proposed Solution

Migrate the embedding storage and similarity search to a dedicated vector database (candidates: Pinecone, Qdrant, or Weaviate). Postgres remains the source of truth for relational data (user accounts, permissions, order history). The vector database only stores embeddings and item IDs, which are joined back to Postgres at query time.

Decision

Approved by engineering leadership (Priya Chandrasekaran, Head of Product, and Marcus Webb, VP Engineering) on March 4, 2026. Target migration completion: end of Q1. Wallace Okafor will lead implementation with support from the platform team.

Risks Noted at Approval Time

Reviewers flagged that moving embeddings out of Postgres could complicate permission-aware filtering, since row-level security policies currently gate which items a user can see in recommendations. The team agreed to monitor this closely after rollout.